

Module 6: Monitoring Resource Use

Research Computing
UNIVERSITY OF COLORADO BOULDER

Be Boulder.

Learning Objectives

- Calculate your resource usage/consumption
- View average wait times in the CURC queues
- Find your relative "priority”
- Check the efficiency of your research workflows

Overview

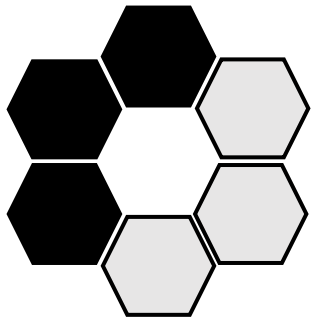
1

2

3

Overview

1



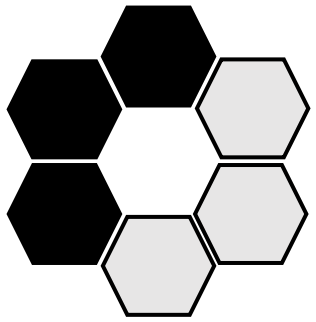
Job Metrics

2

3

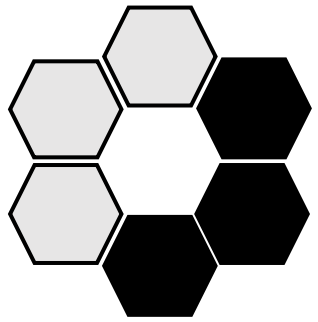
Overview

1



Job Metrics

2

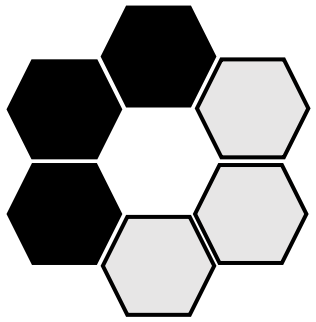


Job Priority

3

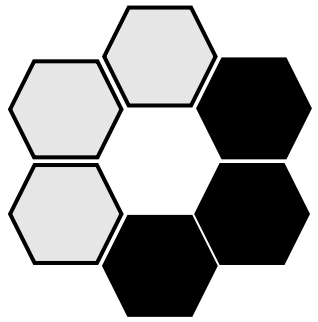
Overview

1



Job Metrics

2

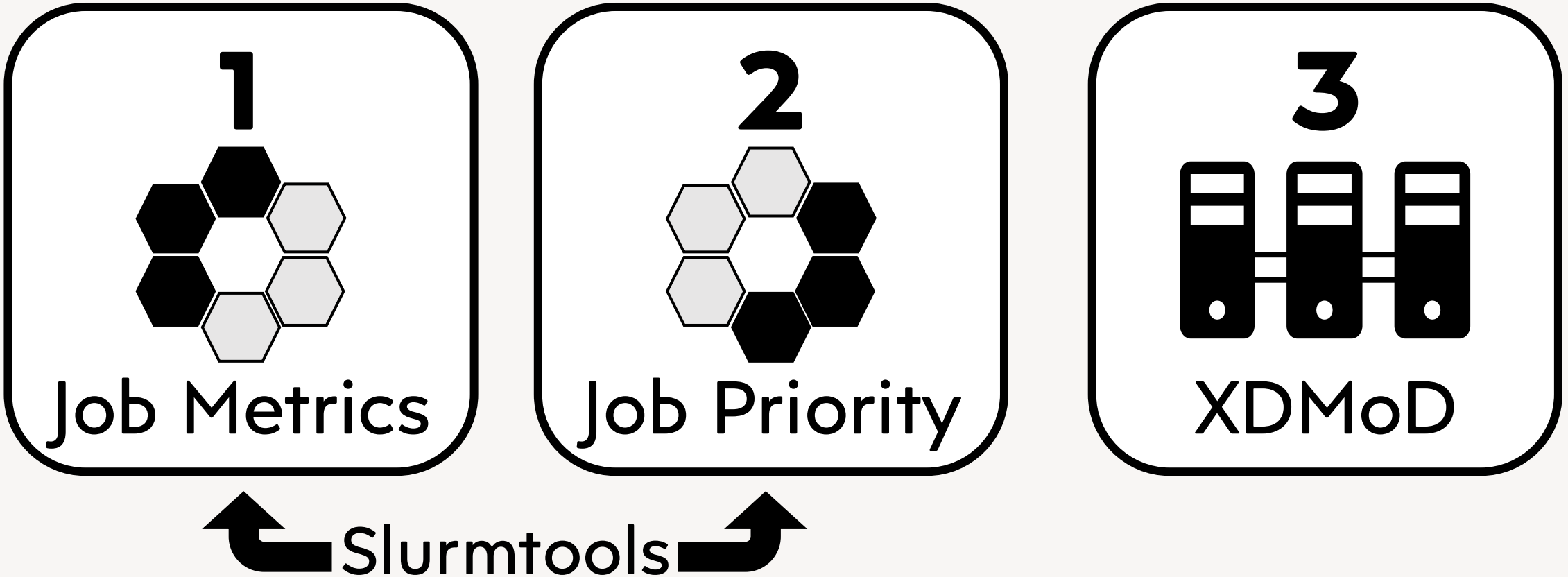


Job Priority

3

Slurmtools

Overview



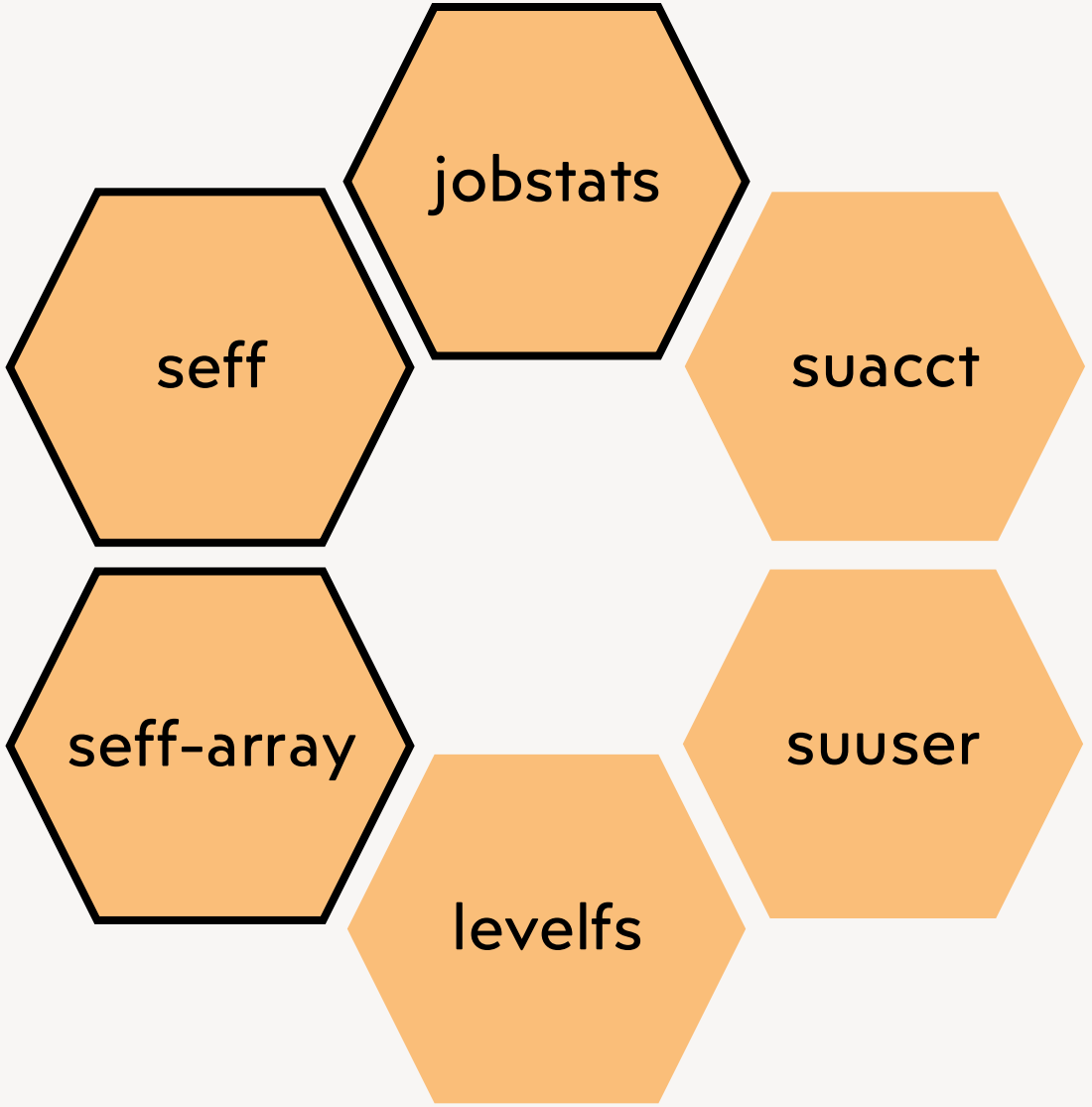
The slurmtools Module

A module that loads a collection of functions to assess recent usage statistics

```
$ module load slurmtools
```

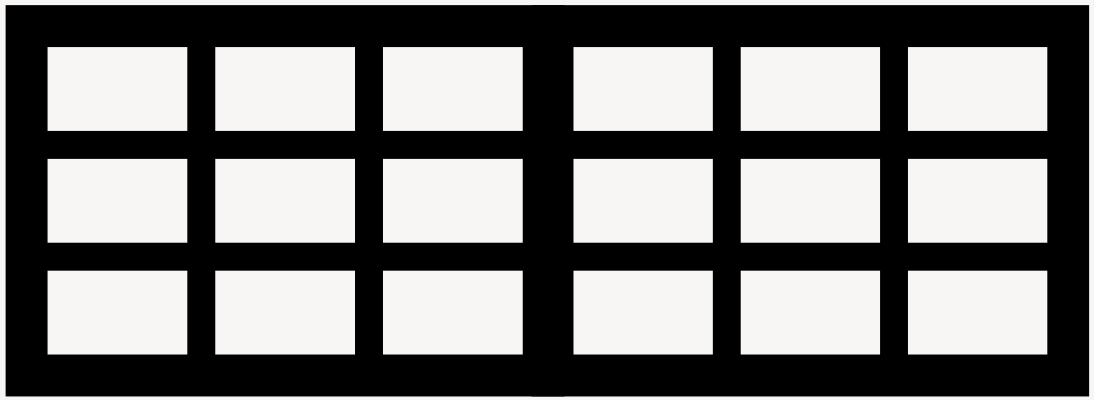


Job Metrics



jobstats

```
$ jobstats $USER <# Days>
```



- jobid
- jobname
- partition
- qos
- account
- cpus
- state
- start-date-time
- elapsed
- wait

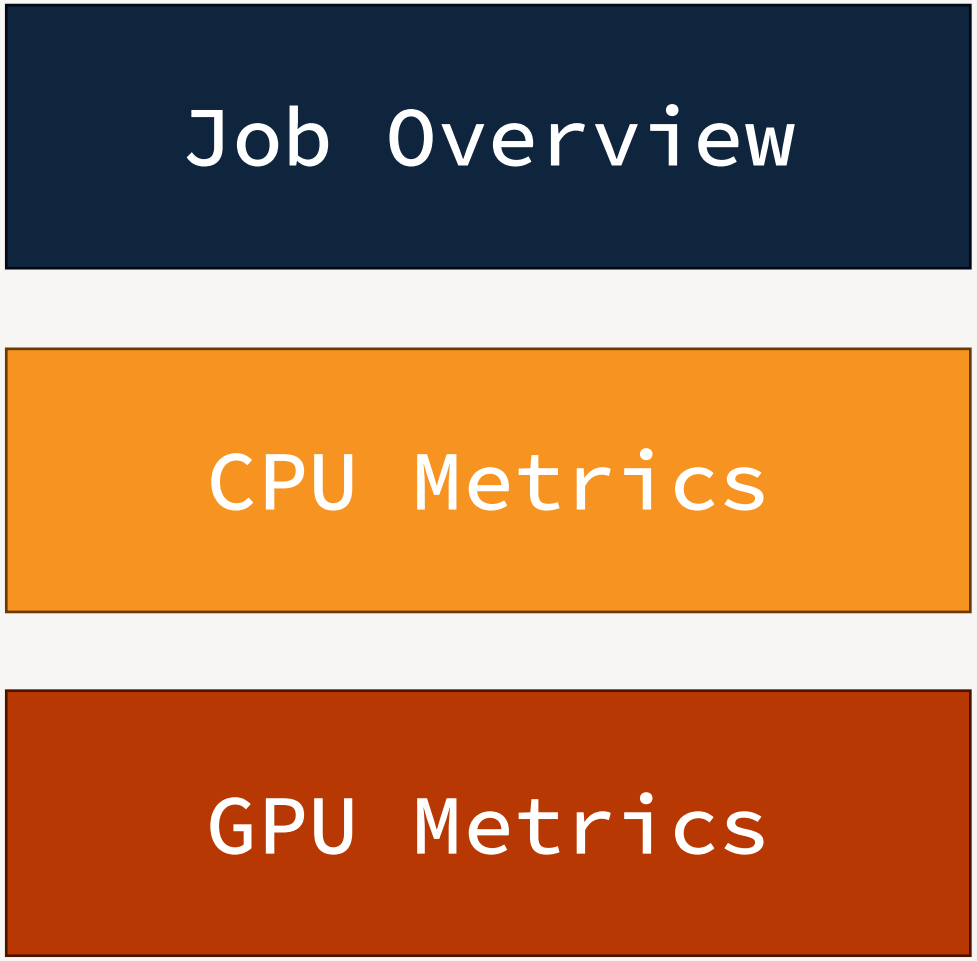
jobstats

```
$ jobstats $USER <# Days>
```

```
job stats for user ralphie over past 7 days
jobid      jobname   partition  qos         account     cpus  state    start-date-time      elapsed     wait
-------------------------------------------------------------------------------------------------------------
28452404   sinterac  acpu       cpu-normal  ucb-gener+  1     CD       2026-06-17T10:43:48  00:00:05    0 hrs
28555838   run_mod+  aa100      gpu-long    ucb-gener+  4     FAILED   2026-06-18T15:51:04  3-00:12:43  84 hrs
```

seff

```
$ seff <jobid>
```



seff

```
$ seff <jobid>
```

Job Overview

Job ID: 20522520

Cluster: alpine

User/Group: ralphie/ralphiegrp

State: COMPLETED (exit code 0)

Nodes: 1 | Cores per node: 21

seff

```
$ seff <jobid>
```

CPU Metrics

CPU Metrics

CPU Utilized: 7-14:11:13

CPU Efficiency: 75.77% of 10-00:27:21 core-walltime

Job Wall-clock time: 11:27:01

Memory Utilized: 21.25 GiB

Memory Efficiency: 27.96% of 76 GiB (76 GiB/node)

seff

```
$ seff <jobid>
```

Did you use the requested CPUs?

CPU Metrics

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Job Wall-clock time: 11:27:01

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CPU Metrics

Did you use the requested CPUs?

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Job Wall-clock time: 11:27:01

Did you use the requested RAM?

Memory Utilized: 21.25 GiB

Memory Efficiency: 27.96% of 76 GiB (76 GiB/node)

seff

```
$ seff <jobid>
```

GPU Metrics

GPU Metrics

Number of GPUs: 1

GPU Type: l40

NOTE: GPU metric availability may vary by GPU type.

Please refer to our documentation for details: [...]

Max GPU Utilization: 41%

Max GPU Memory Utilized: 874.00 MiB

seff

```
$ seff <jobid>
```

GPU Metrics

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seff

```
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```

GPU Metrics

GPU Metrics

Number of GPUs: 1

GPU Type: l40

NOTE: GPU metric availability may vary by GPU type.

Please refer to our documentation for details: [...]

Did you use the requested GPU(s)?

Max GPU Utilization: 41%

Did you pick the *right size* GPU?

Max GPU Memory Utilized: 874.00 MiB

seff-array

```
$ seff-array <jobid>
```

seff-array

Allocated Resources Per Job

```
$ seff-array <jobid>
```

```
Job Information
ID: 12345678
Name: job_array.sh
Cluster: alpine
User/Group: ralphie/ralphiepgrp
Requested CPUs: 1 cores on 1 node(s)
Requested Memory: 1.88G
Requested Time: 00:10:00
```

seff-array

```
$ seff-array <jobid>
```

```
Job Status
COMPLETED: 3
---------------------------------
Finished Job Statistics
Average CPU Efficiency 0.10%
Average Memory Usage 0.00G
Average Run-time 42.00s
```

Allocated Resources Per Job

Average Job Efficiency

seff-array

```
$ seff-array <jobid>
```

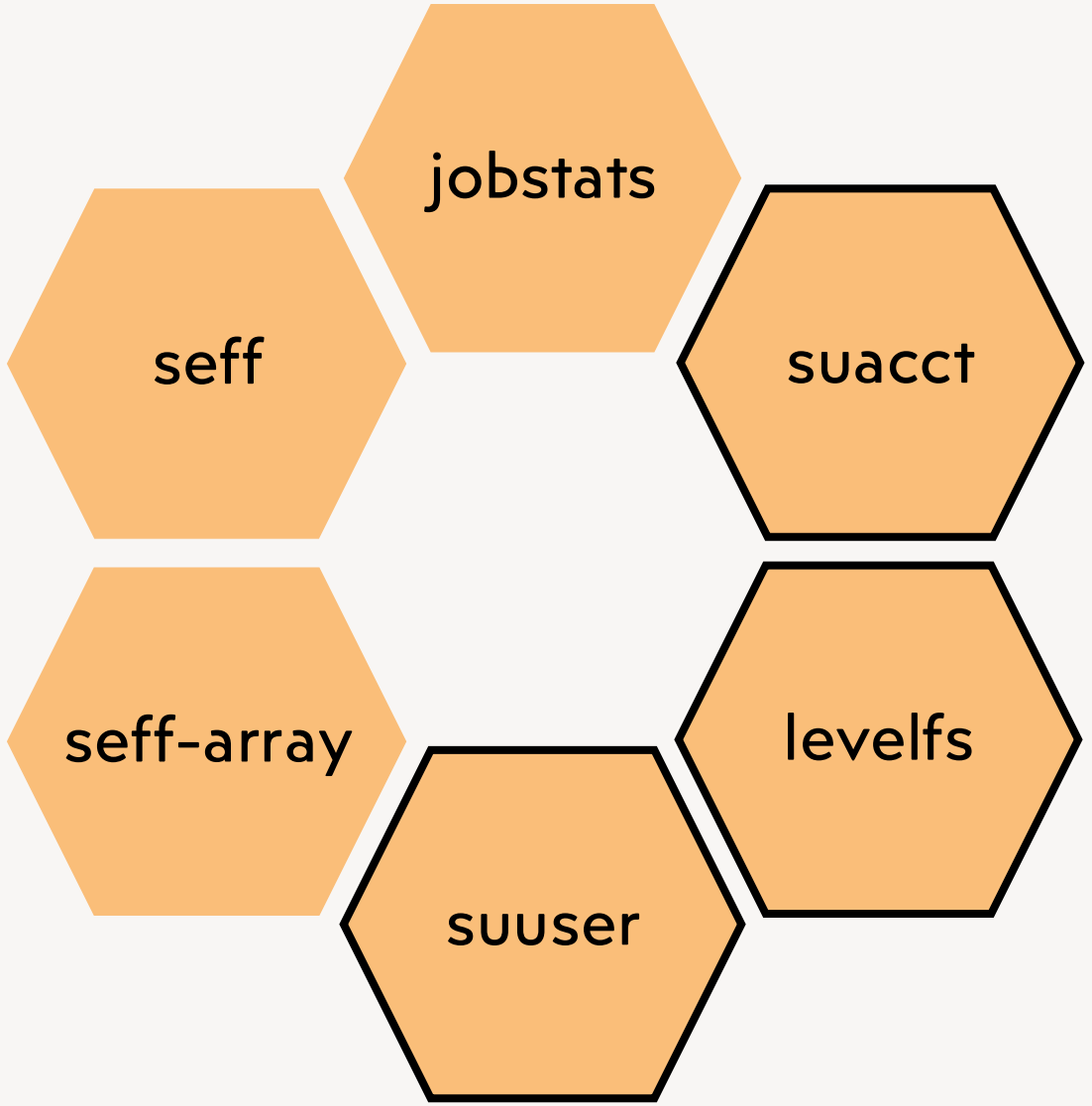
```
CPU Efficiency (%)
--------------------
+0.00e+00 - +1.00e+01   [2] ▇▇▇▇▇▇
+1.00e+01 - +2.00e+01   [1] ▇▇
+2.00e+01 - +3.00e+01   [0]
[…]
+8.00e+01 - +9.00e+01   [0]
+9.00e+01 - +1.00e+02   [0]
```

Allocated Resources Per Job

Average Job Efficiency

Histogram of CPU Efficiency

Job Metrics



Job Priority

Priority

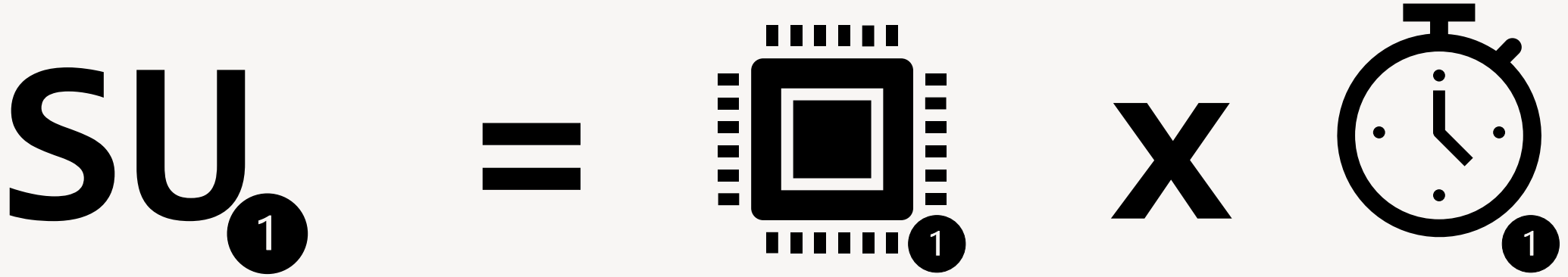
- A job's priority determines its position in the queue.
- A job's priority is based on multiple factors, including:
  - ❑ Age
  - ❑ QOS
  - ❑ Job size
  - ❑ Resources requested
  - ❑ FairShare score

Priority

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  - ❑ Age
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  - ❑ Job size
  - ❑ Resources requested
  - ❑ FairShare score



Service Units



Service units (SUs), sometimes called “core hours”, reflect the processing that a core performs in one hour modified by some scaling factor

suuser

How many Service Units (SUs) have I used?

```
$ suuser <username>
```

```
SU used by user ralphie in the last 30 days:
Cluster |Account     |Login   |Proper Name|TRES Name|Used|
alpine  |ucb-general|ralphie|R. Buffalo |billing  |283 |
```

How do Service Units Impact Priority?

How do Service Units Impact Priority?

Level FairShare

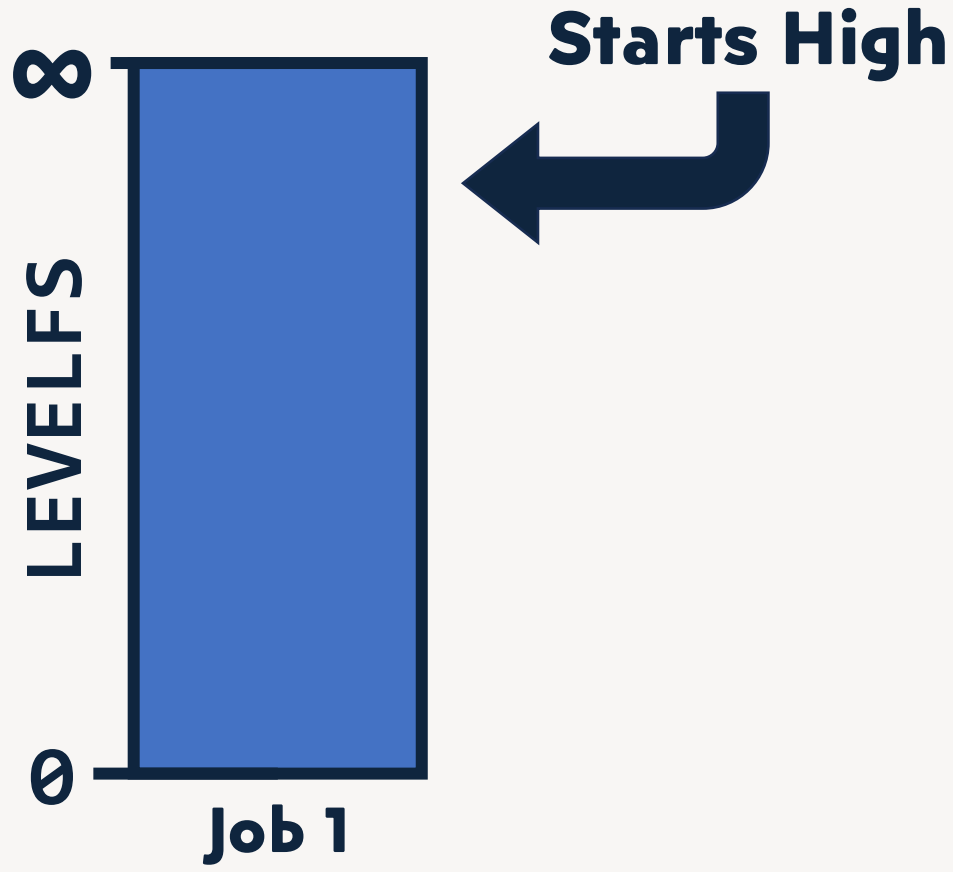
Level FairShare Calculation

$$LevelFS = \frac{Shares}{Usage}$$

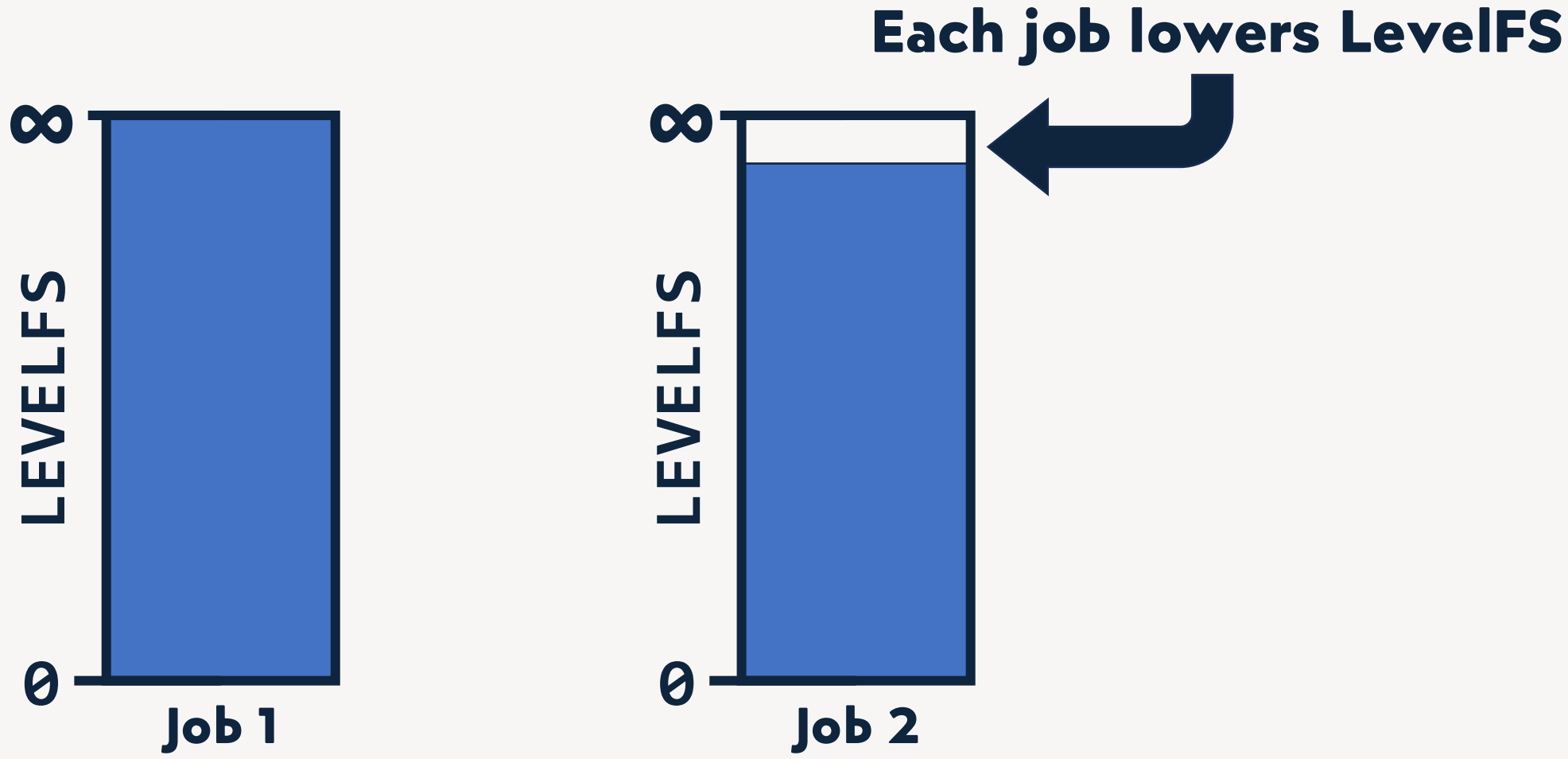
Level FairShare Calculation

$$\text{LevelFS} = \frac{\text{SU Allocated}}{\text{SU Spent}}$$

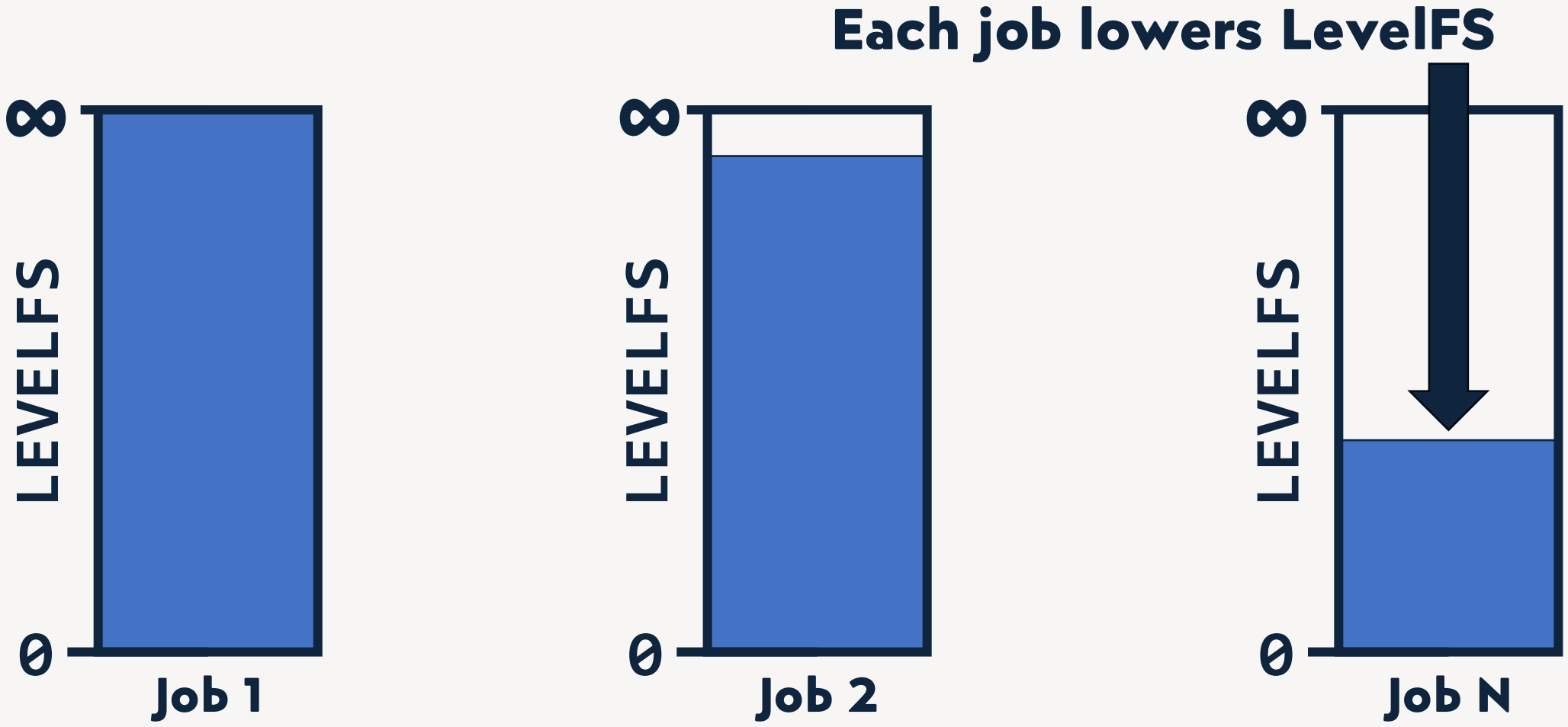
Level FairShare



Level FairShare



Level FairShare



levelfs

```
$ levelfs <username>
```

```
LevelFS for user ralphie and institution ucb:

Account              LevelFS_User         LevelFS_Inst
-------------------------------------------------------
ucb-general          2.015951             0.943096
```

> 1

High Priority

< 1

Low Priority

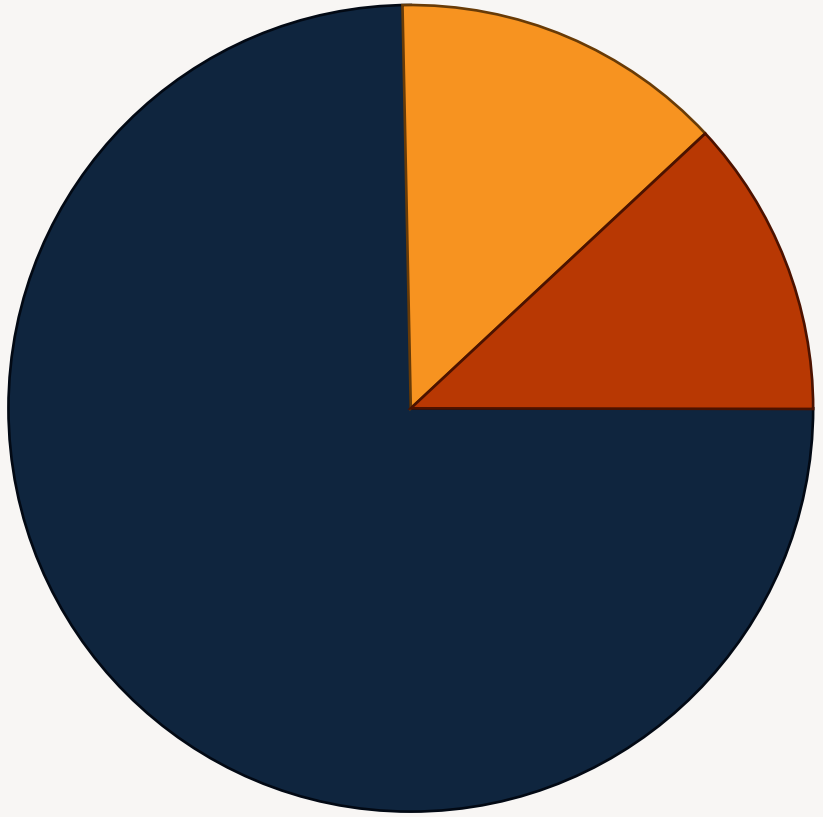
How can I keep my Level FairShare High?

How can I keep my Level FairShare High?

Alpine Allocations

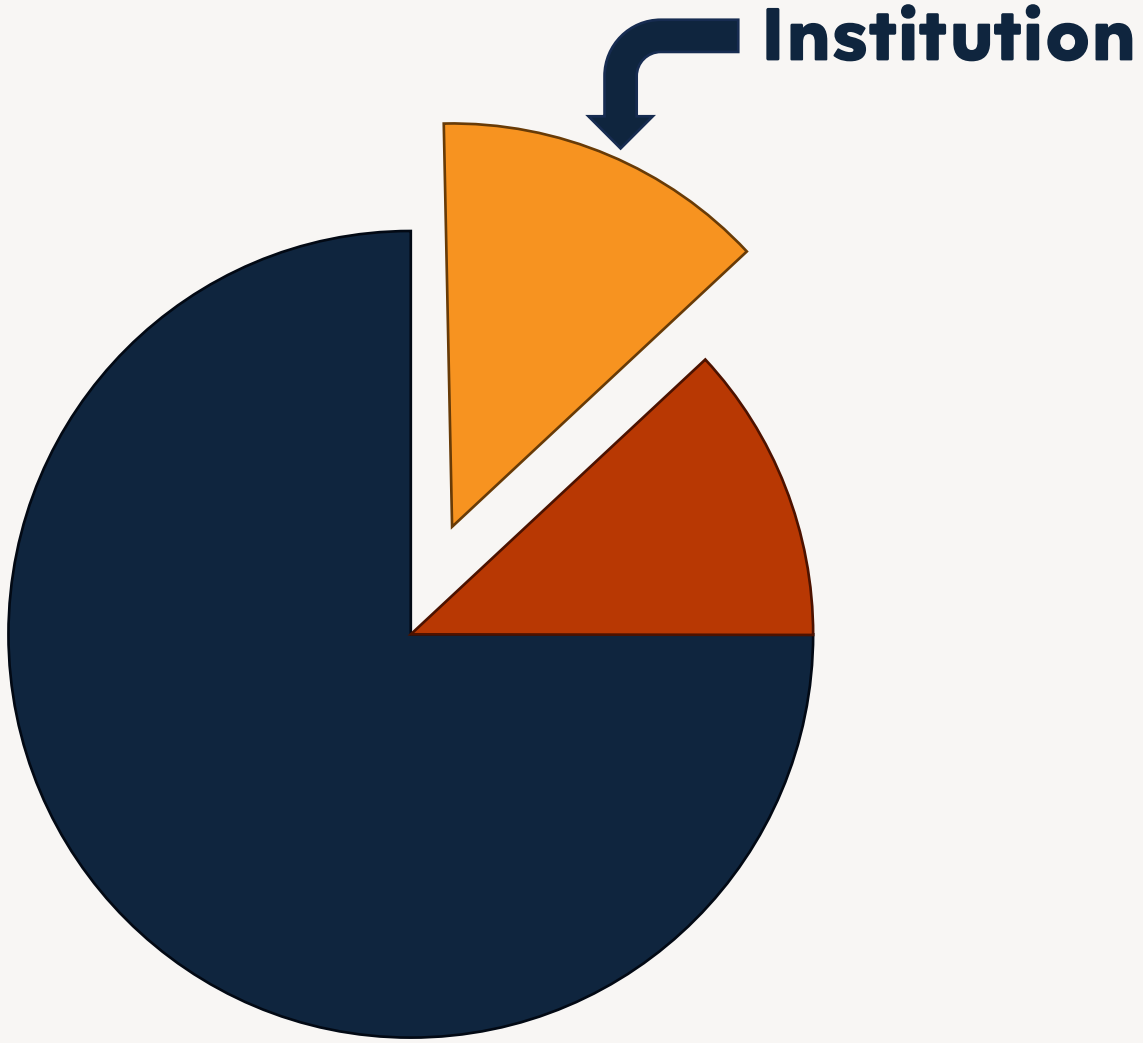
Allocations

An allocation is a way for us to specify your cut of Alpine's computational resources via Service Units.



Allocations

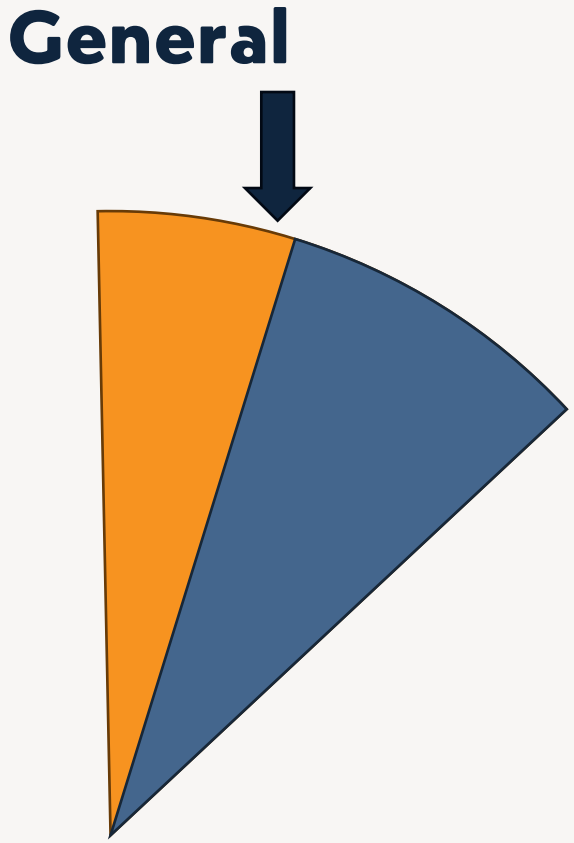
An allocation is a way for us to specify your cut of Alpine's computational resources via Service Units.



Allocations

Each institution has two primary allocations:

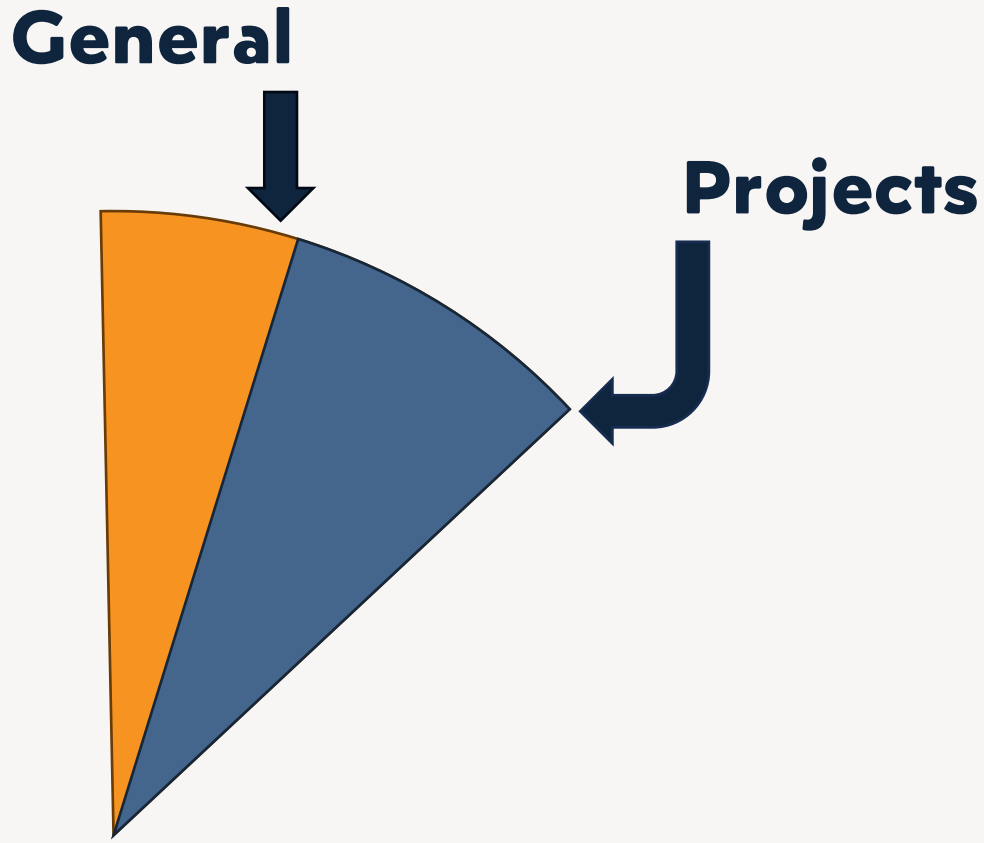
general



Allocations

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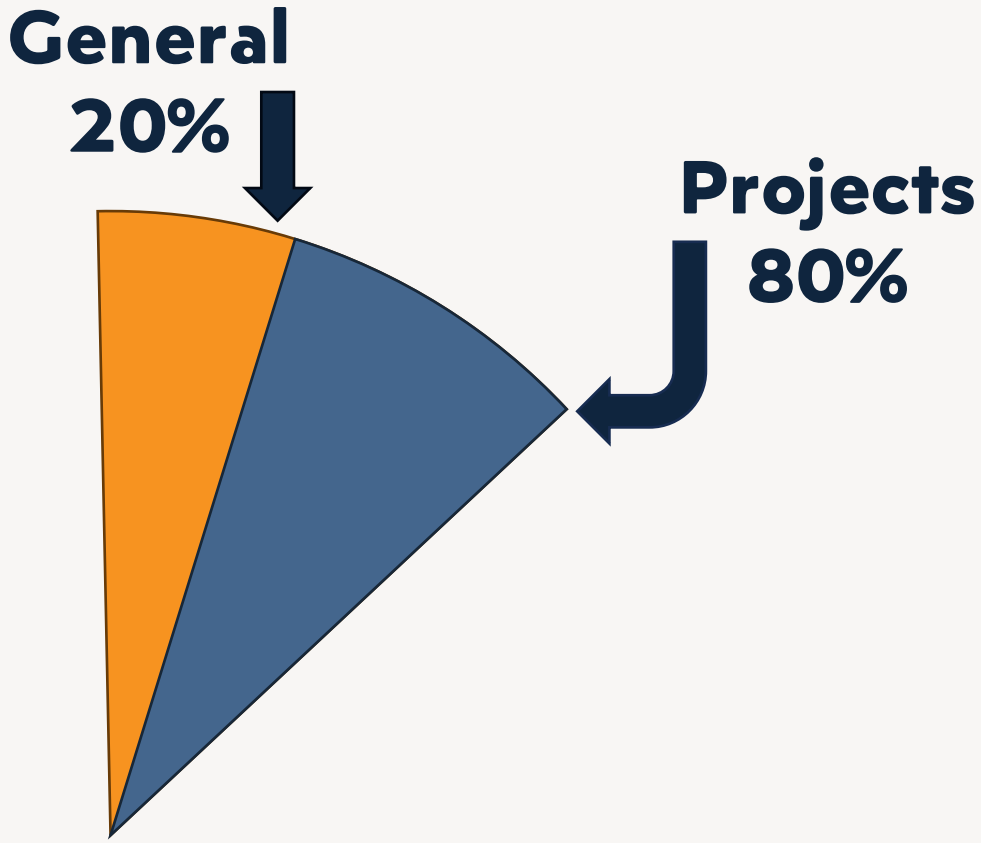
general & projects



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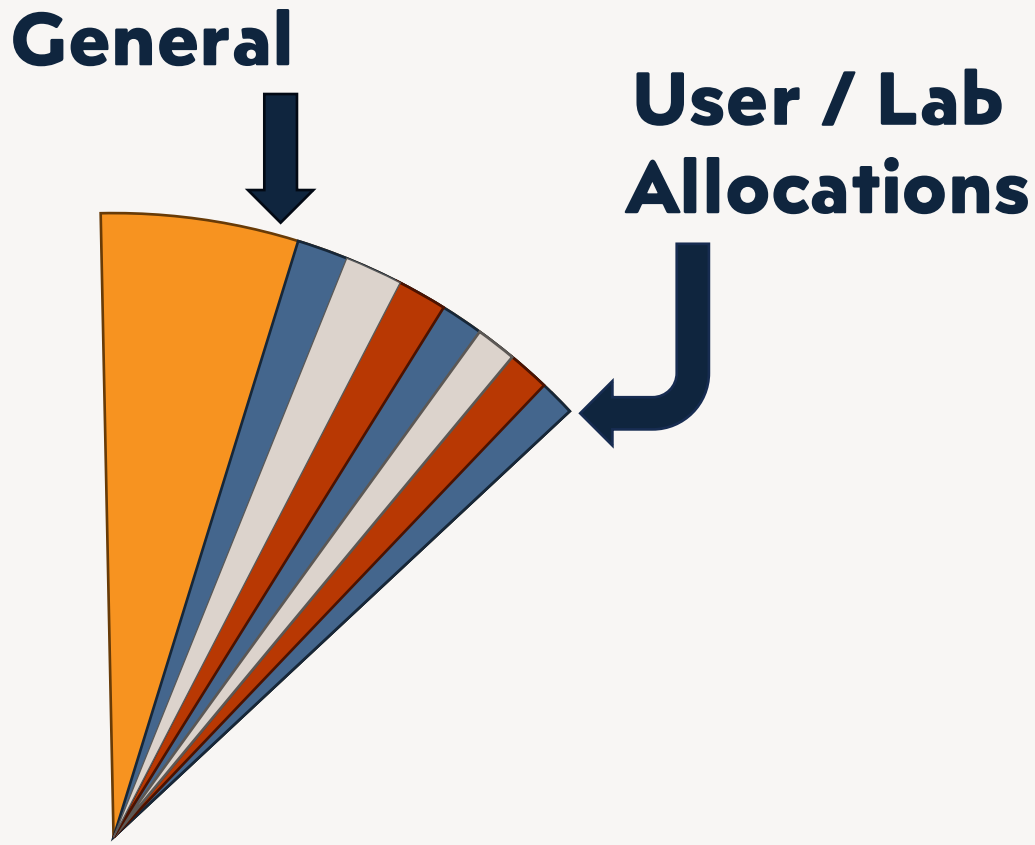
general & projects



Allocations

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general & projects

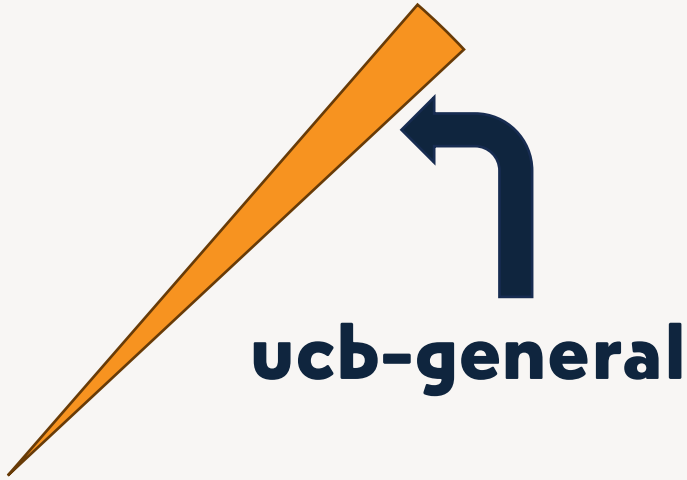


UCB Allocations

Allocation	# of SU's	Approval Process

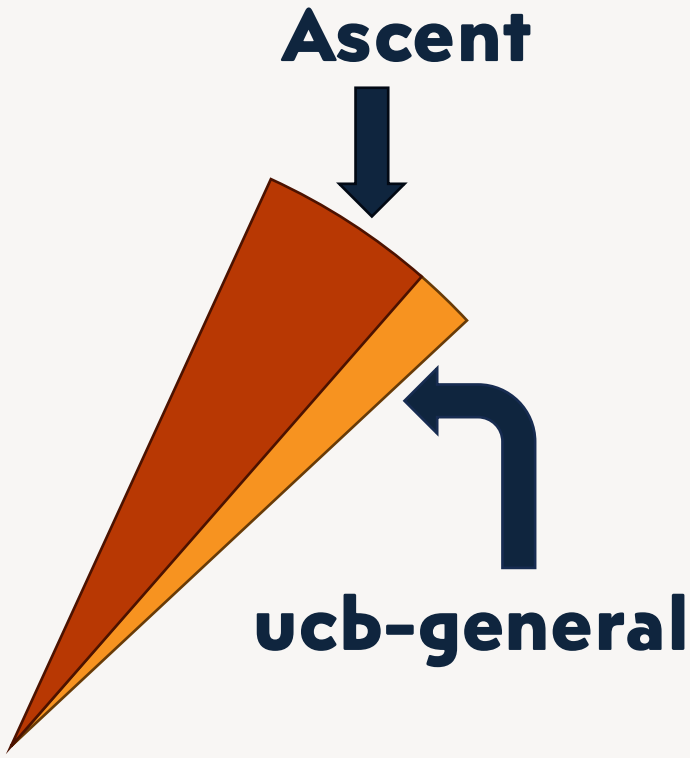
UCB Allocations

Allocation	# of SU's	Approval Process
ucb-general	~3,000	Automatic



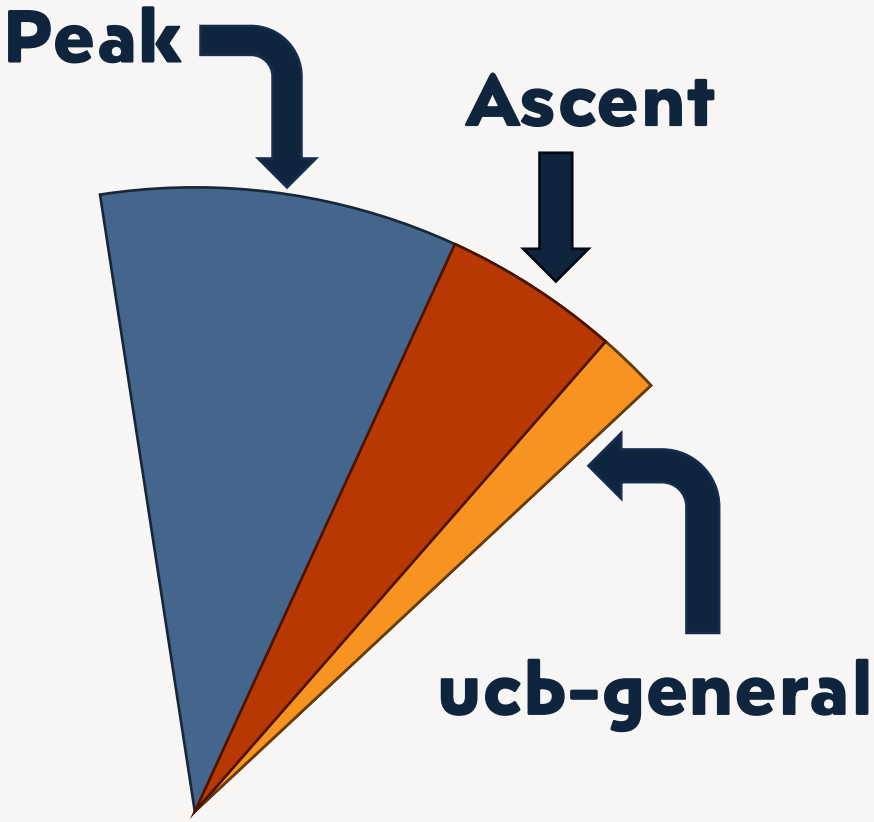
UCB Allocations

Allocation	# of SU's	Approval Process
ucb-general	~3,000	Automatic
Ascent	350,000	Online Form

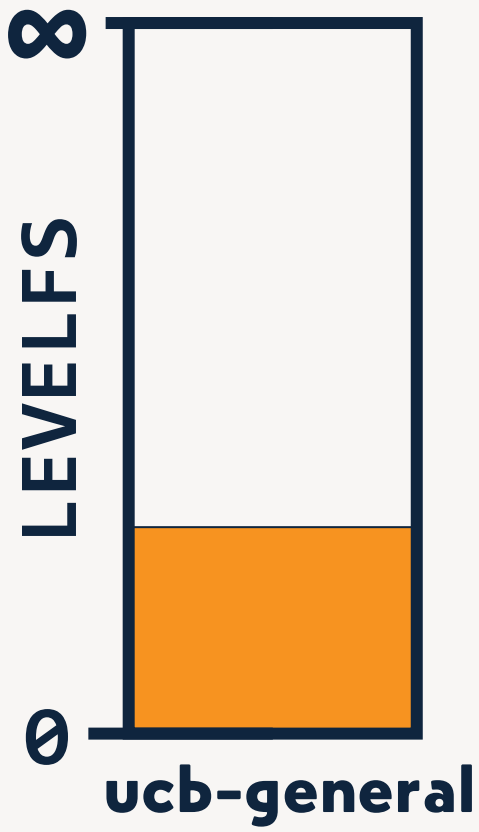


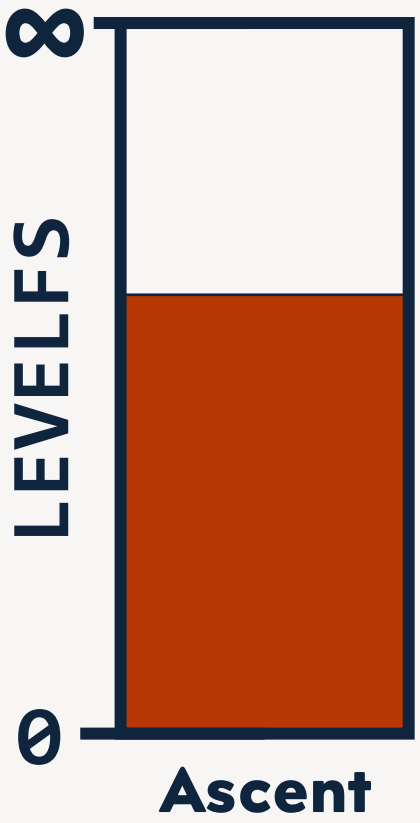
UCB Allocations

Allocation	# of SU's	Approval Process
ucb-general	~3,000	Automatic
Ascent	350,000	Online Form
Peak	350K – 6 Million	Online Form Workflow Review



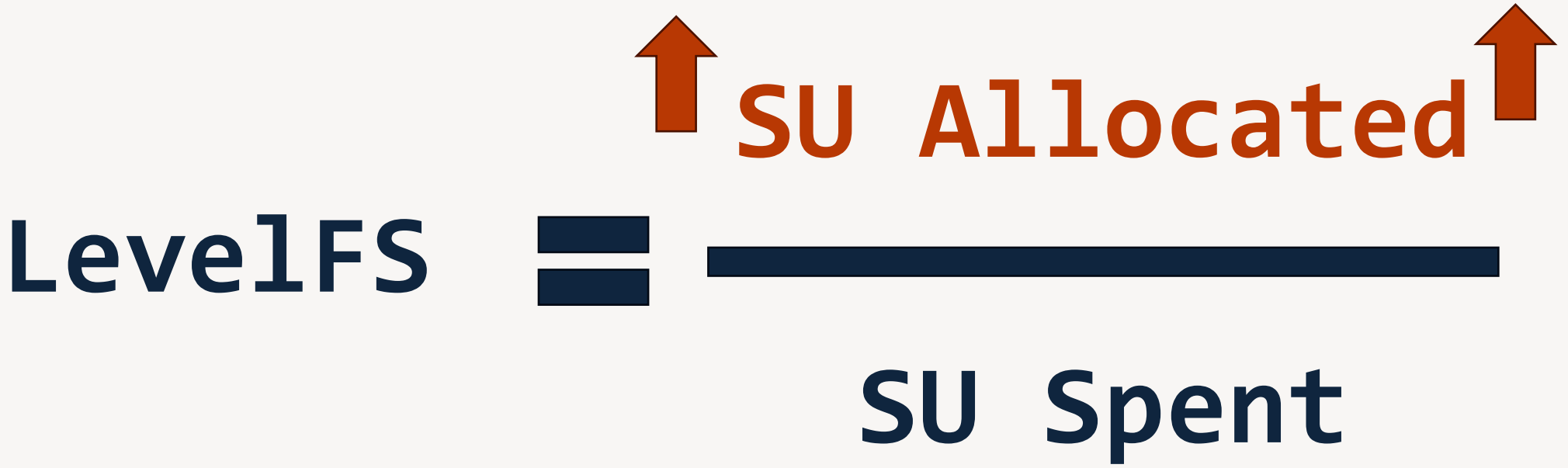
Allocation's Impact on Priority







Level FairShare Calculation



Using Your Allocation

Allocations are referred to as accounts in Slurm's documentation and are indicated by the --account directive:

```
#SBATCH --account=<your allocation name>
```

```
sbatch --account <your allocation name> yourjob.sh
```

suacct

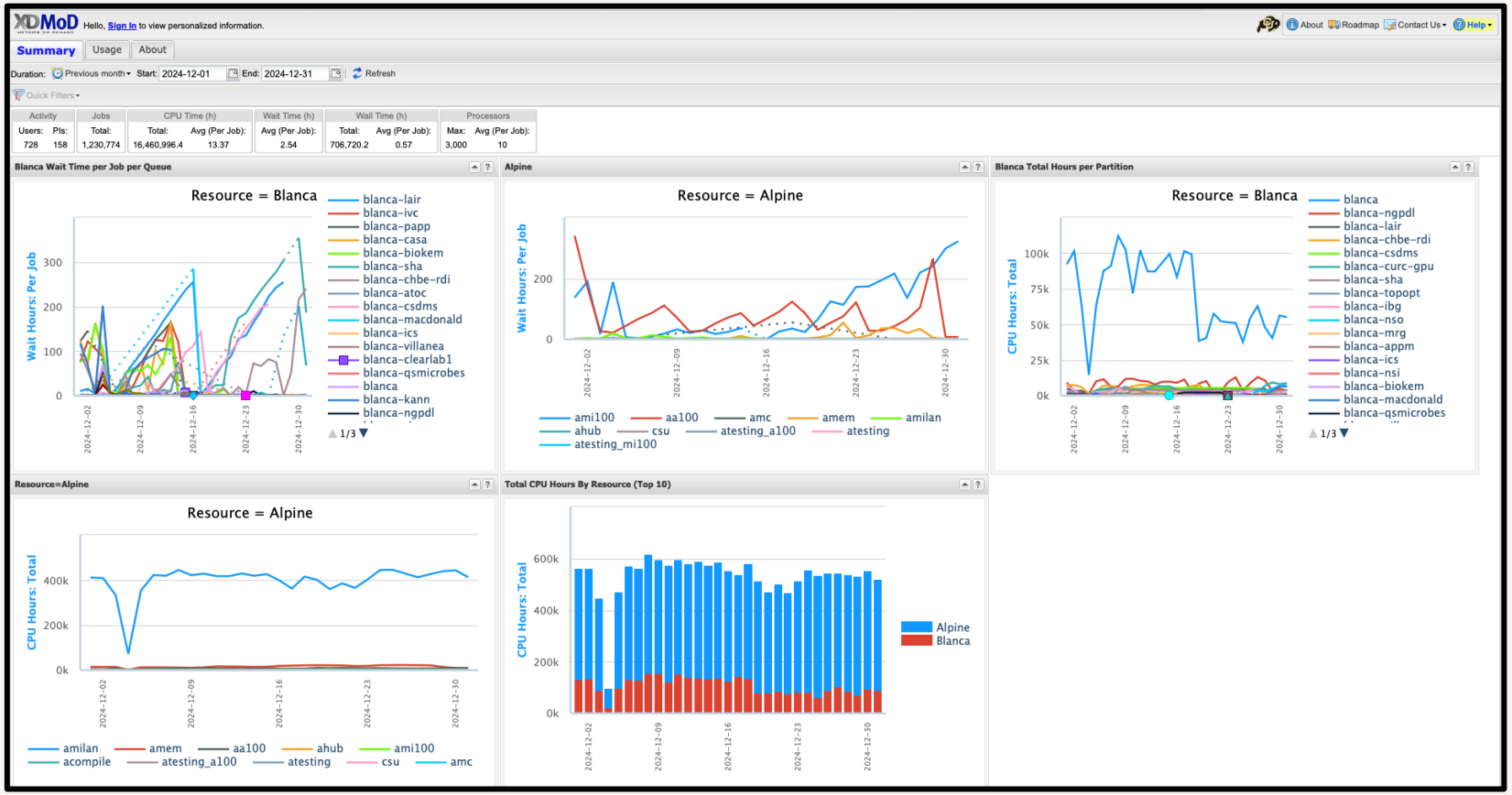
Who is using the SUs on my group's account?

```
$ suacct <account / allocation>
```

```
SU used by account ucb123_asc1 in the last 30 days:
Cluster|Account    |Login  |Proper Name|TRES Name|Used|
alpine |ucb123_asc1|       |           |billing  |5802|
alpine |ucb123_asc1|ralphie|R. Buffalo |billing  |5802|
```

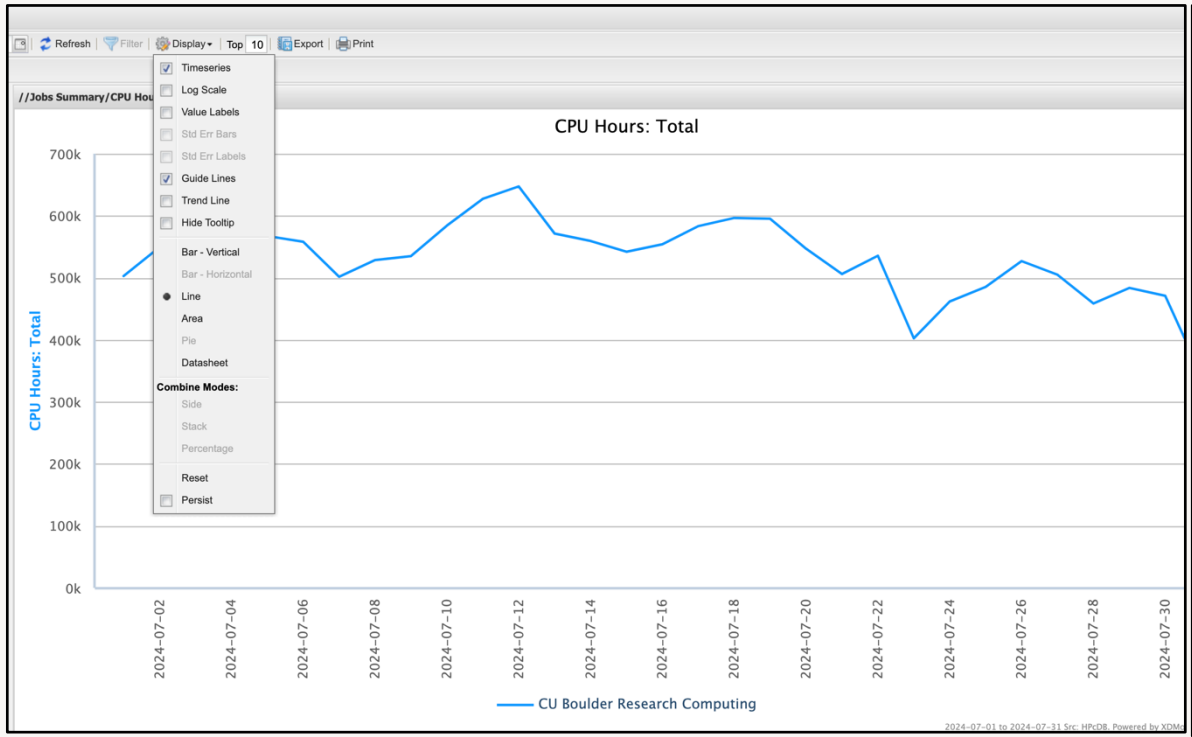

NSF
XDMoD
METRICS ON DEMAND

XDMoD

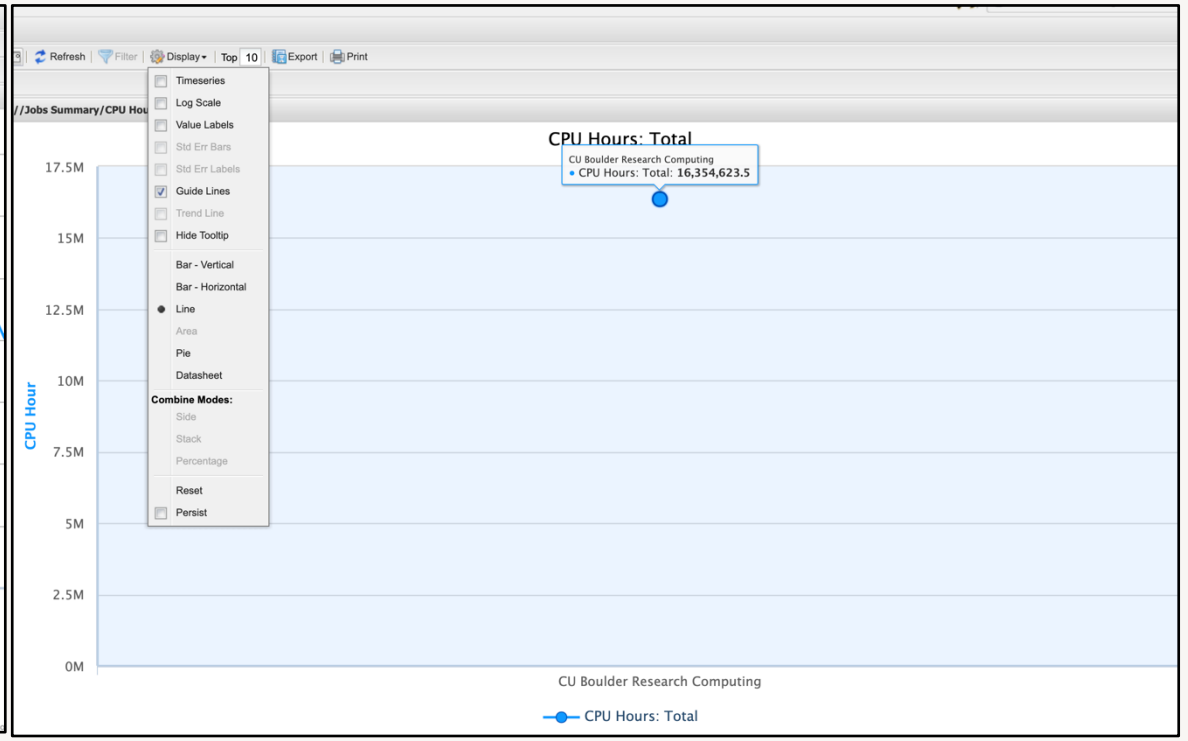


https://xdmod.rc.colorado.edu/

XDMoD Display Menu Defaults

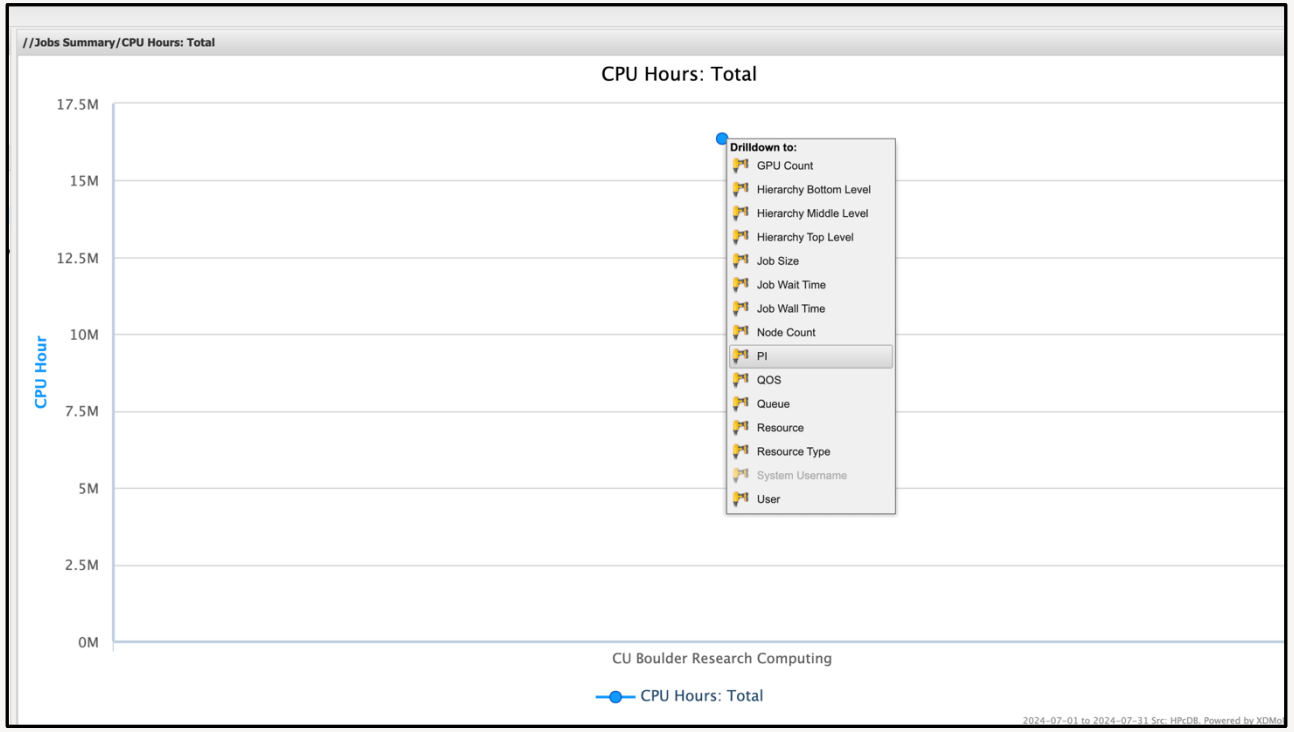


Display Menu: timeseries checked (default)

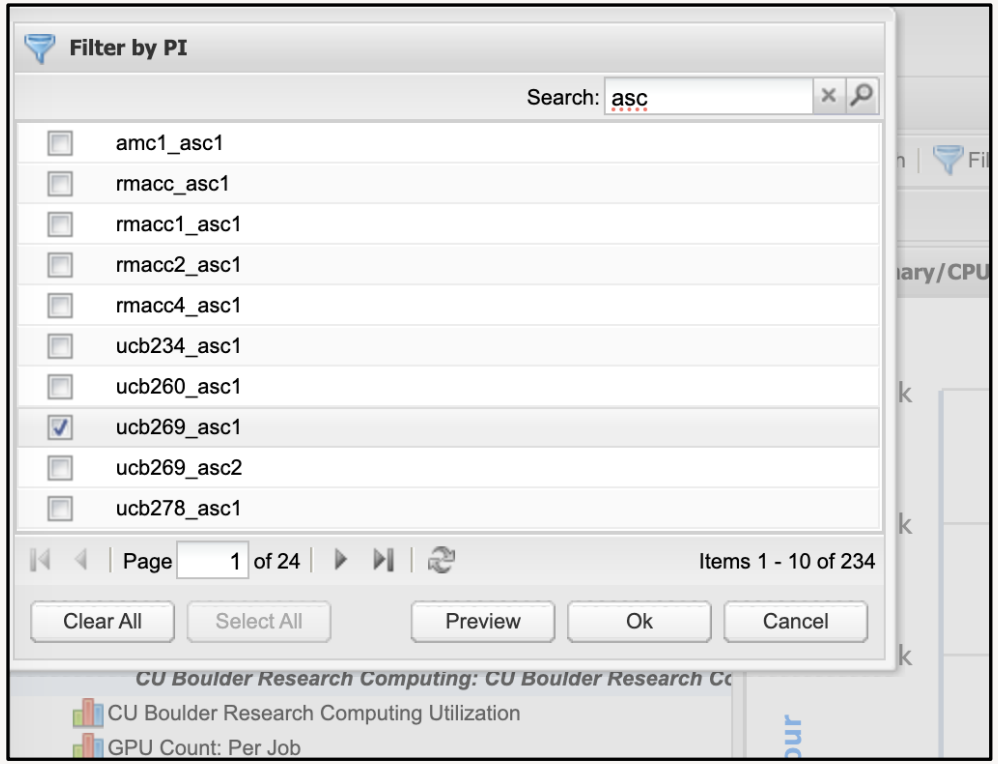


Display Menu: timeseries unchecked

XDMoD Drilldown and Filter



Click on a data point to see Drilldown menu
(PI = Slurm allocation)



Filter and search

Feedback Survey



https://s.zoom.us/m/bPlPdNq5M